

# **Differential Location Analysis of Transcription Factors**

## **-Progress Report 1-**

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### **Goal**

My primary goal is to reveal underlying correlation among different transcription factors in a quantitative way by implementing differential location analysis of ChIP-on-Chip combined with other datasets like Microarray, Protein-Protein Interaction, and Motif analysis. I also aim at characterizing primary and secondary responses to environmental changes which can not be revealed by Microarray experiments alone.

### **What I have done**

1. Primary Data Handling
  - a. Data acquisition
    - i. ChIP-on-Chip data: 203 yeast transcription factors in 12 different conditions from the Young Lab at MIT
    - ii. Gene expression data from SMD (Stanford Microarray Database): Only two highly similar experimental condition heat-shock and oxidative stress ( $H_2O_2$ ) to ChIP-on-Chip experiments.
    - iii. Protein-Protein Interaction data between 203 TF from BIND.
  - b. Data pre-processing
    - i. ChIP-on-Chip data (ratio and p-value) was pre-processed by removing null data and combining p-value and ratio data
2. Correlation Analysis
  - a. Correlation of ratio between different conditions for every available pairwise condition for different p-value thresholds.
  - b. Highly differentially bound genes were obtained and compared with previously found TF bound list (Young et al)
3. Statistical Analysis
  - a. In order to confirm the distribution of the highly differentially bound genes is not following normal distribution and rather skewed, Chi-Square tests have been performed against normal distribution.
4. Sequence Analysis
  - a. Each promoter regions were scanned for possible TFBS via MATCH program against TRANSFAC databases.

### **Results so far**

1. Highly differentially bound genes showed a skewed pattern to either positive or negative direction away from normal distribution which implies they are significantly different from being random.
2. I was not able to deduce reliable quantitative relationship between different factors from the comparison of gene expression microarray experiment and ChIP-on-Chip analysis.
3. We found some clues which can be used to differentiate primary response from secondary response of gene expression to environmental changes. Such cases are genes whose gene expression level did not changed but their binding affinity dramatically changed.

### **Remaining works**

1. Refining cases of primary and secondary responses.
2. Adding supporting biological evident for this refinement. (Literature searches)
3. Functional analyses of primary and secondary response genes